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SECTOR AT A GLANCE

Steel Sector

Manufacturing in India

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Sector at a Glance

Steel Sector

Manufacturing

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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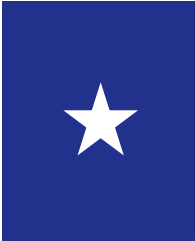
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1. Overview



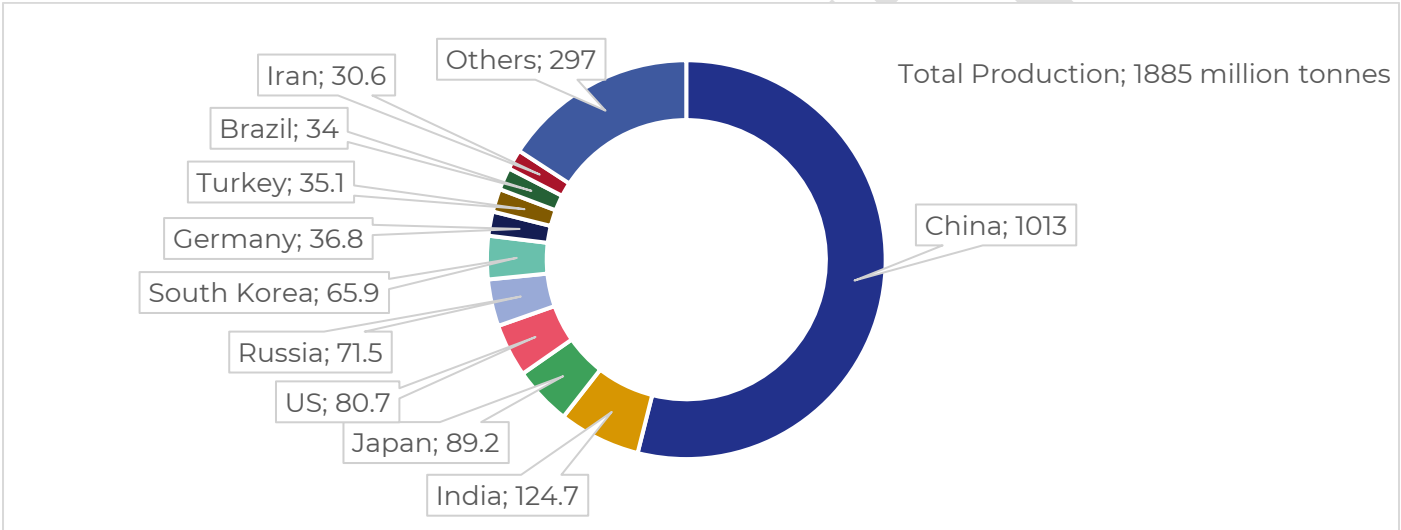
India is the second-largest producer of crude steel worldwide, accounting for 7% of global production in 2022.¹

In FY23, production rose 5.7% on-year to 127.2 million tonnes.²

Steel has a 1.7% share of India's gross domestic product. It has an output multiplier effect of 1.4x and employment multiplier effect of 6.8x.^{3,4}

A. Overview of the global and domestic steel industries

Figure 1 - Global crude steel production, 2022



Source: World Steel Association 1; Unit: Million tonnes (MT)

India’s growing demand, geographical location and long coastal boundary make it a key player in the global steel market. Supported by ample iron ore availability and favourable policies, domestic steel production and capacity have increased steadily over the past few years, propelling India to become the second-largest producer in 2018, overtaking Japan.

In 2022, global crude steel production stood at 1,885 million tonnes. Of this, India’s share was ~7%.¹ India’s crude steel production increased from 101.4 million tonnes in 2017 to 125.4 million tonnes in 2023.¹ In 2021 and 2022, India registered double-digit growth in demand, well supported on the supply side through capacity additions.

In 2022, global per capita apparent consumption of steel averaged 222 kg. Meanwhile, India’s per capita apparent consumption of steel was close to 81 kg, highlighting huge growth potential.² To harness the domestic potential, the Government of India launched the National Steel Policy, 2017, with a vision to create a technologically advanced and globally competitive steel industry and increase domestic consumption to 160 kg by FY31.³

1 World Steel Association, 2023 World Steel in Figures

2 Ministry of Steel, Joint Plant Committee, Annual statistics 2016-17, 2017-18, 2021-22, 2022-23

3 Ministry of Statistics and Programme Implementation, National Accounts Statistics 2023

4 Ministry of Steel, National Steel Policy 2017

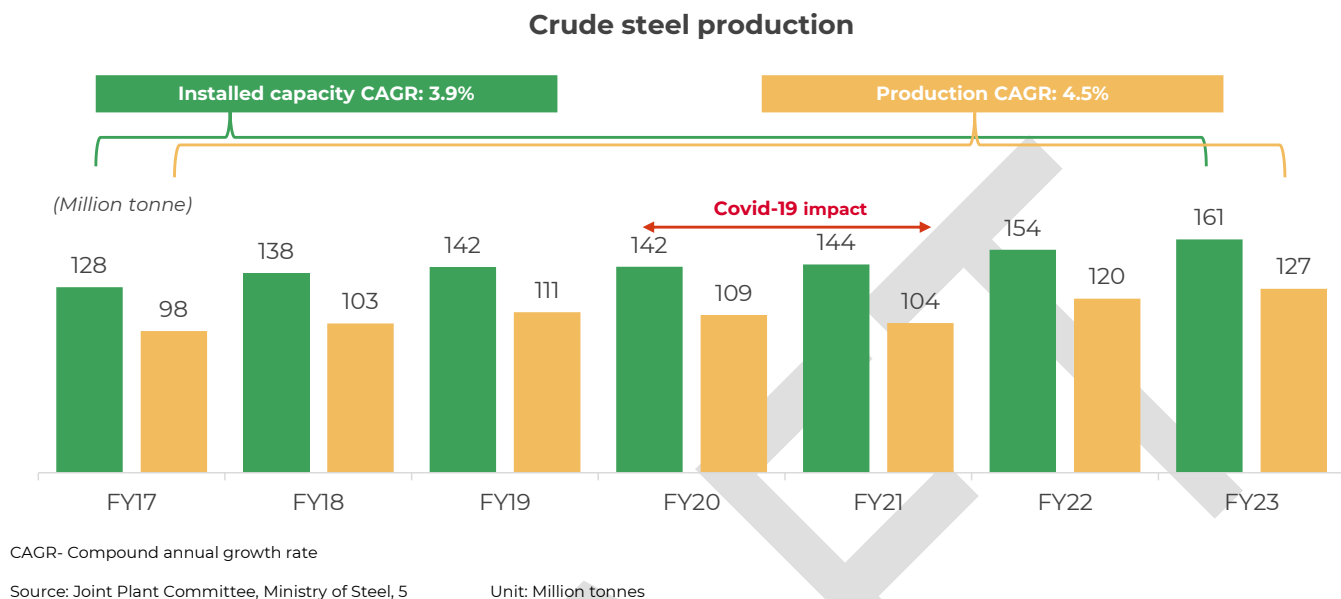
5 Ministry of Steel, Joint Plant Committee, Annual statistics 2016-17, 2017-18, 2021-22, 2022-23

3 Ministry of Steel, National Steel Policy,2017

PRIMARY COLOURS		SECONDARY COLOURS	
	C-70, M-10, Y-80, K-0 R-61, G-161, B-90		C-0, M-60, Y-40, K-0 R-242, G-132, B-130
	C-5, M-30, Y-70, K-0 R-242, G-187, B-94		C-60, M-0, Y-40, K-0 R-105, G-192, B-172
	C-0, M-80, Y-45, K-0 R-234, G-81, B-103		C-00, M-20, Y-70, K-0 R-254, G-208, B-97
	C-45, M-30, Y-0, K-0 R-153, G-170, B-215		C-20, M-0, Y-10, K-0 R-214, G-236, B-235
	C-91, M-61, Y-0, K-0 R-0, G-102, B-178		C-56, M-0, Y-6, K-0 R-84, G-208, B-237

2 Indian steel sector

Figure 2 - Crude steel production in India



A. Methods of steel production

Steel production can be divided into three stages:

- 1) Iron making**, wherein iron ore is reduced from its oxide form using a reducing agent and various other fluxes.
- 2) Steel making** is the next stage in the process, which uses iron produced in the first stage as an input material. Carbon content as well as other trapped impurities are removed, and alloying elements are added to get the desired properties and cast into slabs for flat products and billets or blooms for long products. End-products at this stage are also known as 'semi-finished steel'.
- 3) Downstream processes**, wherein the slabs or billets are further processed in the mills into various finished steel products such as reinforcement bars, hot-rolled coils (HRC), wire rods, galvanised plane sheets (GP), galvanised corrugated (GC) sheets, angles, channels, etc.

- **Iron making**

- Blast furnaces are the most common method of iron production around the world. Blast furnaces produce molten iron from iron ore, with coke made from coking coal acting as the reducing agent and limestone and dolomite being used as flux. Around two-thirds of India's iron is produced by this method.
- Direct reduction of iron ore to iron without melting is another common method of iron making. Thermal coal, natural gas, or syngas (produced by coal gasification) is used as the reducing agent. The product thus formed is called directly reduced iron, or DRI.

- **Steel making**

- Oxygen-based steel making involves the use of pure oxygen to convert a charge of molten iron from a blast furnace into steel. Around 45% of Indian crude steel is produced by this method.

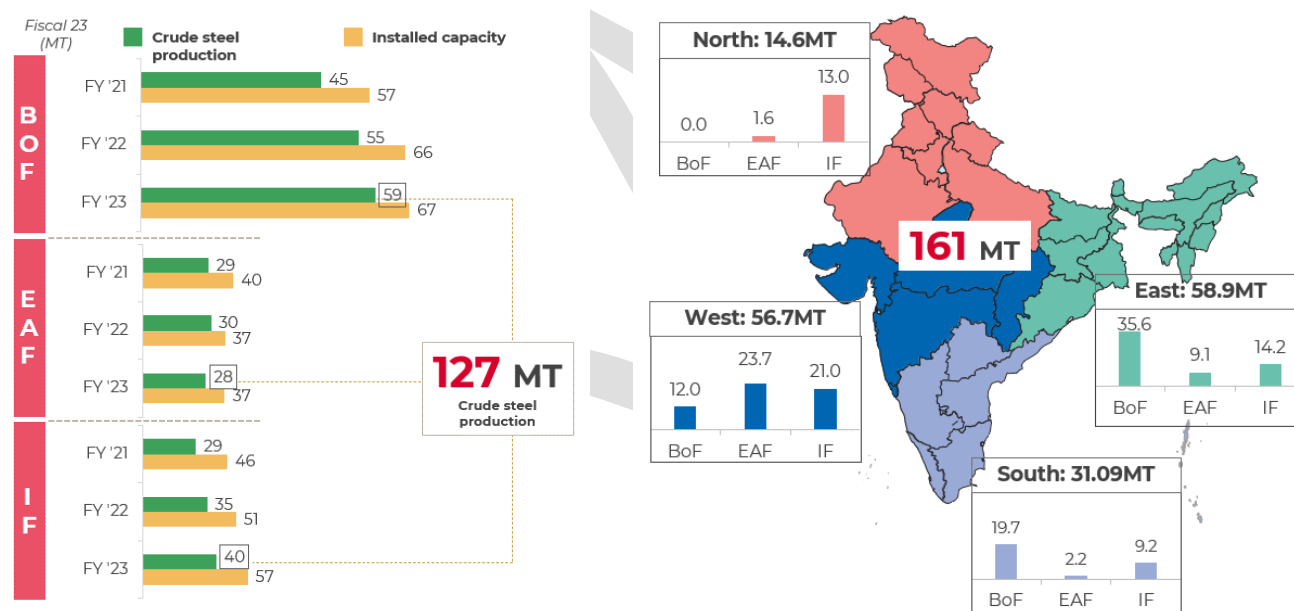
- Steel making using an induction furnace where DRI and steel scrap are melted using electricity is another preferred method of steelmaking in India; about 30% of Indian steel is produced by this method.
- The use of an electric arc furnace or a combination of electric arc and oxygen for steelmaking is another method to produce steel. Steel scrap, DRI, or liquid blast furnace iron are used as major raw materials.

The ability to produce steel on a large scale at a moderate cost makes blast furnace, followed by basic oxygen furnace, a preferred route for steelmaking globally. In India, large-scale steel manufacturing follows the same route, even though it involves importing about 70% of the coking coal requirement.⁴ Lower capital costs and the domestic availability of iron ore and thermal coal have led to the growth of the DRI (induction furnace) route in India.

Steel is 100% recyclable; however, increasing demand and the substitution of concrete with steel make it necessary for new steel to be produced from iron ore globally. Developed economies, which have scrap steel available for recycling, are moving towards meeting the majority of their steel requirements via recycling.

B. Indian steel production

Figure 3 - Installed capacities and crude steel production in India



BoF: Basic oxygen furnace; EAF: Electric arc furnace; IF: Induction furnace
Source: JPC

Source: Joint Plant Committee, Ministry of Steel; 8 Unit: Million tonnes
BOF: Basic oxygen furnace EAF: Electric arc furnace IF: Induction furnace

Of the total crude steel production of 127 million tonnes in India in fiscal 2023, 63% was produced by the six largest steel producers by capacity⁵; 37% was produced by more than 900 manufacturers.⁵ Geographically, Odisha, Jharkhand, Chhattisgarh, Maharashtra, Karnataka, and West Bengal account for about 74% of India's steel production.⁹ The availability of iron ore is a major factor in the development of the steel industry in these states. If produced from iron ore, around three tonnes of raw materials are required to produce one tonne of steel. This makes the availability of logistics infrastructure particularly important for the development of the steel industry.

⁴ Ministry of Coal, Office of Coal Controller, Released July 2022, Provisional coal statistics 2021-22

⁵ Ministry of Steel, Joint Plant Committee, Annual statistics 2022-23

Table 1 – Crude steel capacity and production trend in India

Particulars (million tonnes)	FY19	FY20	FY21	FY22	FY23
Crude steel production	110.9	109.1	103.5	120.3	127.2
Crude steel capacity	142.2	142.3	143.9	154.1	161.3
Capacity Utilisation %	78.0%	76.7%	71.9%	78.1%	78.9%
Finished steel production	101.3	102.6	96.2	113.6	123.2
Import	7.8	6.8	4.8	4.7	6.0
Export	6.4	8.4	10.8	13.5	6.7
Stock changes	4.1	0.9	(4.7)	(1.0)	2.61
Apparent steel consumption	98.7	100.2	94.9	105.8	119.9

Source: Joint Plant Committee, Ministry of Steel ⁹

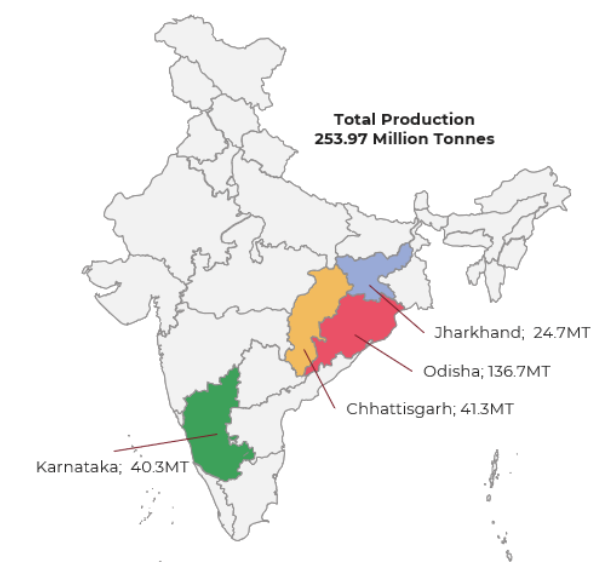
Finished steel can be divided into flat and long categories. While flat steel is produced using steel slabs, long steel is produced by using billets or blooms as rolling mill feed. Of the 123 million tonnes of finished steel produced in FY23, 56 million tonne (46%) was flat steel, and 67 million tonnes (54%) belonged to the long steel category.⁹

C. Raw material for steel production

Raw material costs account for a substantial portion of the total cost of steel production and hence have a significant effect on steel prices. While India has an abundant supply of iron ore, it is largely dependent on imports to meet its coking coal demand. The availability of iron ore supply had been the most dominant factor in the location of ore-based steel making in the country, while steel recyclers are spread across the country based on availability of steel scrap.

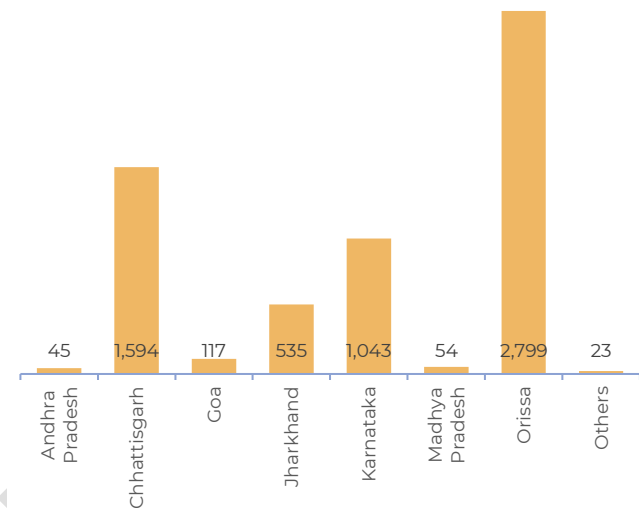
To boost the domestic availability of iron ore, the government of India has not only amended the MMDR Act but has also increased the auctioning of new iron ore mines. Since the amendment of the MMDR Act, more than 100 mines have been auctioned, accounting for 5.2 billion tonnes of iron ore reserves.

Figure 4 - Indian iron ore production for FY22 **Figure 7 - Indian iron ore reserves (MT)**



Source: Indian Bureau of Mines, (6&7)

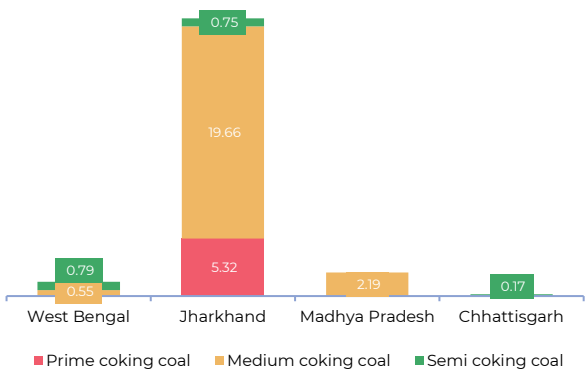
Unit: Million tonnes



Source: Indian Mineral Yearbook, 2021 (8)

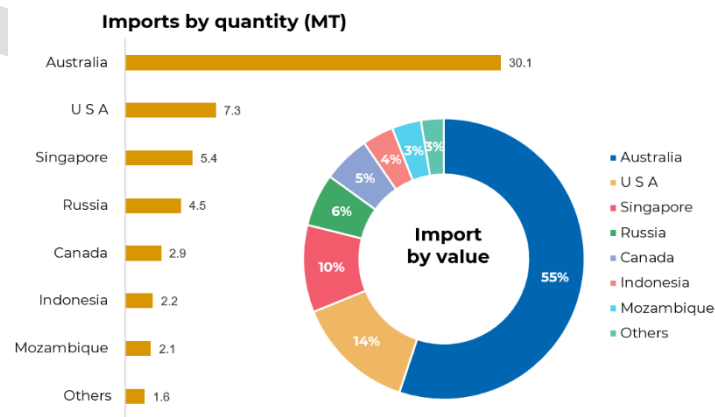
Most of the coking coal found in India has a high ash content and is not suitable for use in the iron-making process directly. Complexities relating to the beneficiation of coking coal to reduce ash content and preferences in coal supply given to power plants make domestic coking coal supply limited. Weather-related disruptions, geopolitical tensions, the Covid-19 pandemic, and resultant supply chain disruptions have significantly impacted raw material prices in the past. In FY23 India's production of washed coking coal was 5.313 MT.⁹

Figure 6 - Indian coking coal reserves (BT)



Source: National Coal Inventory – 2022, CMPDIL 10

Figure 7 - Import of coking coal in India- FY23 (MT)



Source: Directorate General of Foreign Trade, Ministry of Commerce, and Industry, 11

6. Ministry of Mines, Indian Bureau of Mines, Monthly Statistics of Mineral Production, March 2022
7. Ministry of Commerce and Industry, Directorate General of Foreign Trade, Export Import Data Bank
8. Ministry of Mines, Indian Bureau of Mines, Indian Mineral Yearbook, 2021.
9. Office of the coal controller, Ministry of coal, Provisional coal statistics 2023
10. Central Mine Planning & Design Institute Limited, National Coal Inventory – 2022

D. Domestic steel consuming sectors

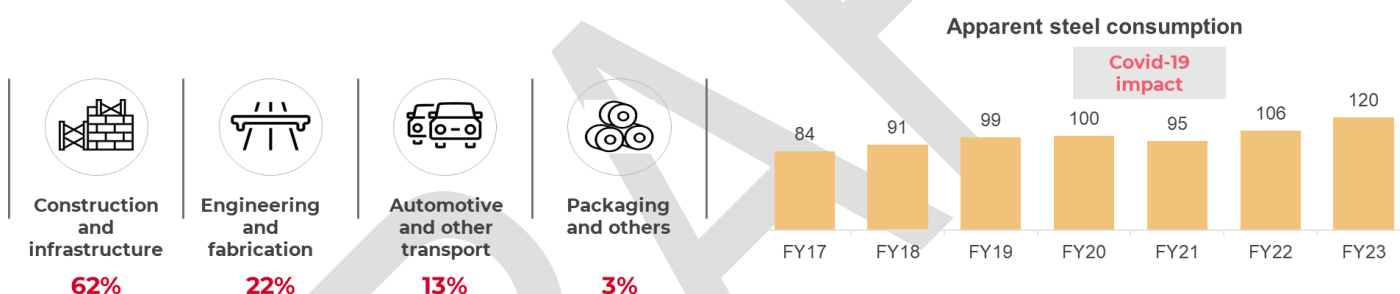
Domestic steel consumption increased 13.3% on-year in fiscal 2023, driven by demand pull from allied sectors, which is a result of increased government spending in steel-heavy sectors.⁹

The key steel consuming sectors in India can be broadly categorised into: construction and infrastructure, engineering and fabrication, automotive and other transport, and packaging and others. More than half of the demand comes from the construction and infrastructure segment (62%) followed by engineering and packaging (22%), while automotive and other transport (13%) and engineering and other segments (3%) have a smaller share of the demand pie.

While majority of long steel demand comes from the building and construction segments, flat steel demand is predominantly from the infrastructure, automobile, engineering and packaging segments.

In FY23, India's apparent consumption of steel was 119.89 million tonnes; of this, 60.4 million tonnes were in non-alloy long steel, 50 million tonnes were in non-alloy flat steel, 6 million tonnes in alloy steel, and 3.4 million tonnes in stainless steel.

Figure 8 –Steel consumption in India



Source: National Steel Policy (11) & Joint Plant Committee (5)

E. Carbon emissions in steel production

The steel sector accounts for 7-9% of global carbon emissions.¹² It is among the six hard-to-abate sectors because carbon is required for the reduction of iron ore. India's emissions of ~2.5 tonne of CO₂ per tonne of crude steel (tCO₂/tcs) are much higher than the global average of 1.91 tCO₂/tcs due to the poor availability of scrap for furnace charge mix in the production of crude steel.^{13 & 14}

Steel produced via the induction furnace (IF) and electric arc furnace (EAF) routes has a higher share of sponge iron, which is more polluting than the traditional blast furnace-basic oxygen furnace (BF-BoF) route. Therefore, even with a higher share of crude steel production via the electric routes as compared with Europe and Japan, India's emission levels are much higher.

The steel scrap recycling policy and the vehicle scrappage policy, with the aim of increasing the availability of steel scrap, are crucial for reducing emissions and the fight against climate change.

Additionally, Perform, Achieve and Trade (PAT), a flagship scheme under the National Mission for Enhanced Energy Efficiency Mechanism, which is a market-based mechanism wherein energy-saving

¹¹ Ministry of Steel, National Steel Policy-2017

¹² World steel association, Public policy paper: Climate change and the production of iron and steel, 2021

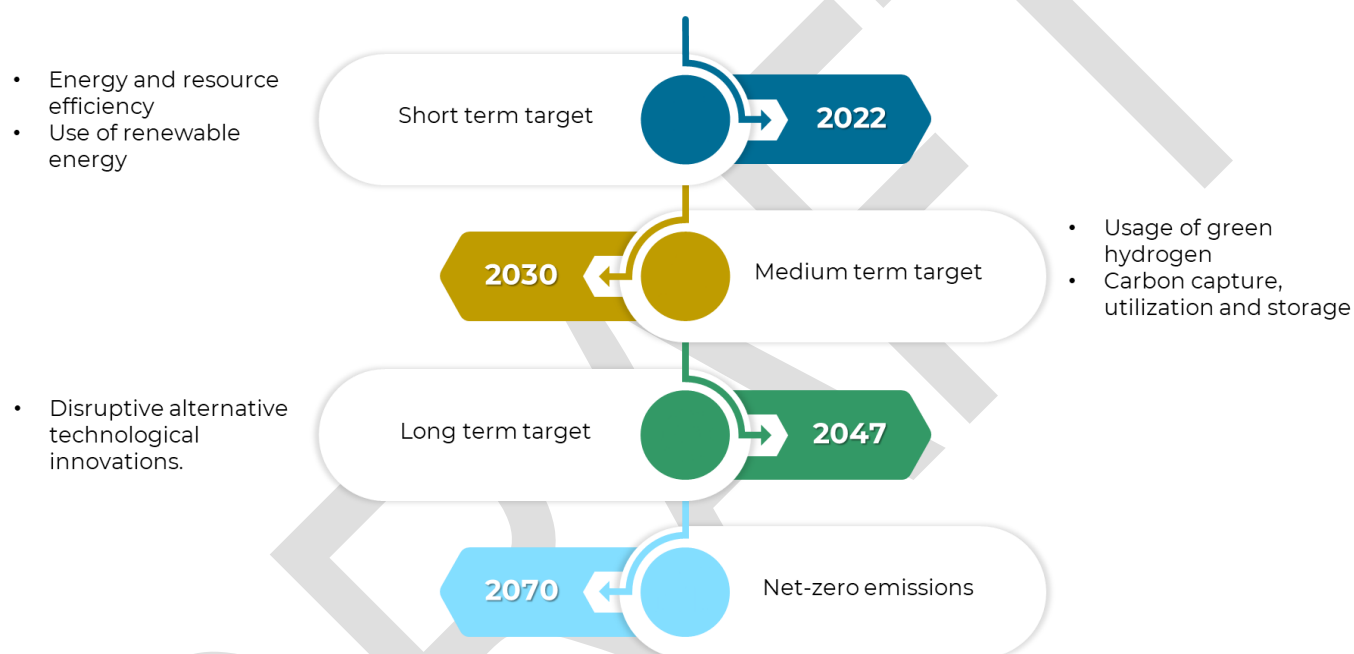
¹³ Ministry of steel, Energy & environment management in steel sector

¹⁴ World Steel Association, Released 2022, Sustainability Indicators 2022 report

certifications are tradable, was launched in 2012. The iron and steel sector successfully reduced 2.1 million ToE (tonne of oil equivalent) in the first cycle of PAT and 2.9 million ToE in the second cycle.¹⁵ Six more cycles of PAT have been notified, which will further reduce energy consumption.

Ministry of steel has adopted India's targets to achieve carbon neutrality by year 2070. The process is divided into three terms: short, medium & long term. This will enable smooth implementation of initiative and government support in the direction as per the phases of implementation. During the first phase, the ministry has also invested in research & development of technologies for the next phase of implementation through 'National Green Hydrogen Mission'¹⁶ & various other research grant schemes such as 'Promotion of Research & Development in Iron & Steel Sector'.¹⁷

Figure 9 – Ministry of Steel's Net-Zero target



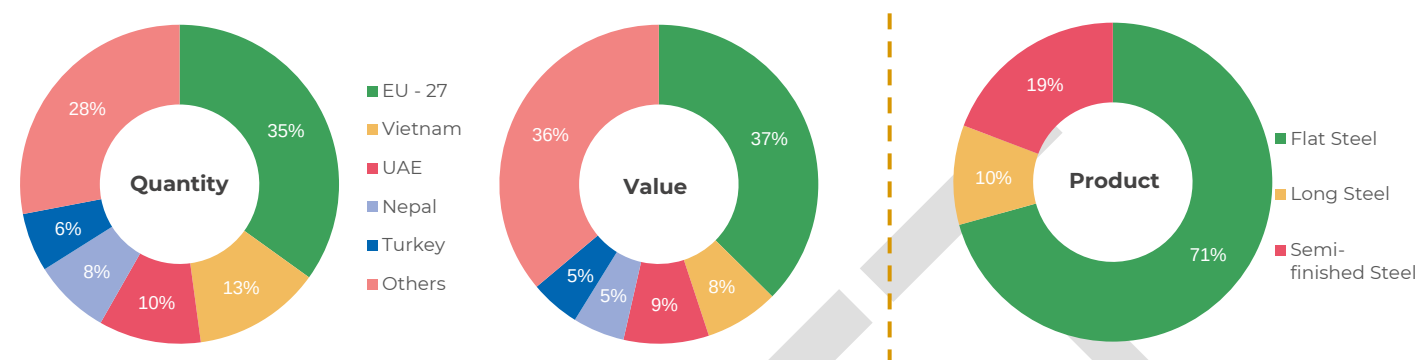
¹⁵ Ministry of Steel, 2023, Annual Report 2022-23

¹⁶ Ministry of steel, Invitation of R&D proposals under National Green Hydrogen Mission, May 2023

¹⁷ 2023, Ministry of steel, R&D scheme, Promotion of Research & Development in Iron & Steel Sector.

3. Steel trade scenario

Figure 10 - Finished steel exports - FY23



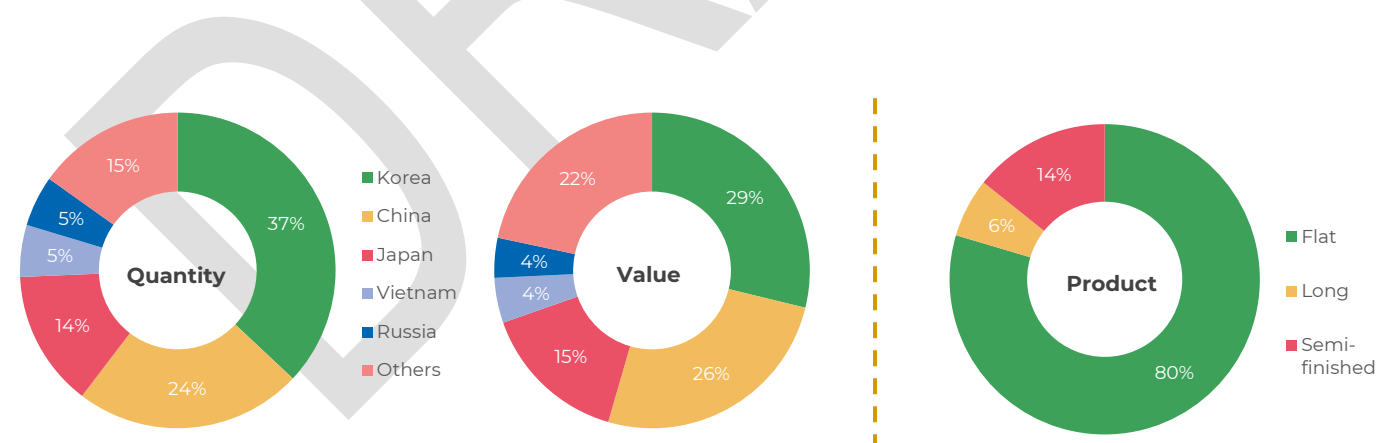
Source: Joint Plant Committee, Ministry of Steel, 9

Unit: Million tonnes

India has been a net exporter of steel. Iron ore availability has added to the country's cost advantage, keeping domestic mills competitive in the steel market. Volume- and value-wise, the EU-27 remains the largest export destination. While volume-wise, Vietnam is the 2nd largest importer of Indian steel, value-wise, the UAE occupies the 2nd position as of fiscal 23.⁹

However, the imposition of trade and traffic barriers by importing nations to promote adoption of greener ways of steel production and protection against cheaper imports remains a key monitorable. In line with the same, the US government imposed a 25% tariff on steel imports in 2018, which led to a significant decline in exports to the US. Similarly, the European Union (EU) is in the process of implementing the Carbon Border Adjustment Mechanism (CBAM), which will increase the cost of exports to Europe, a key export destination for Indian steel players.

Figure 11 - Finished steel imports - FY23



Source: Joint Plant Committee, Ministry of Steel, 9

Unit: Million tonnes

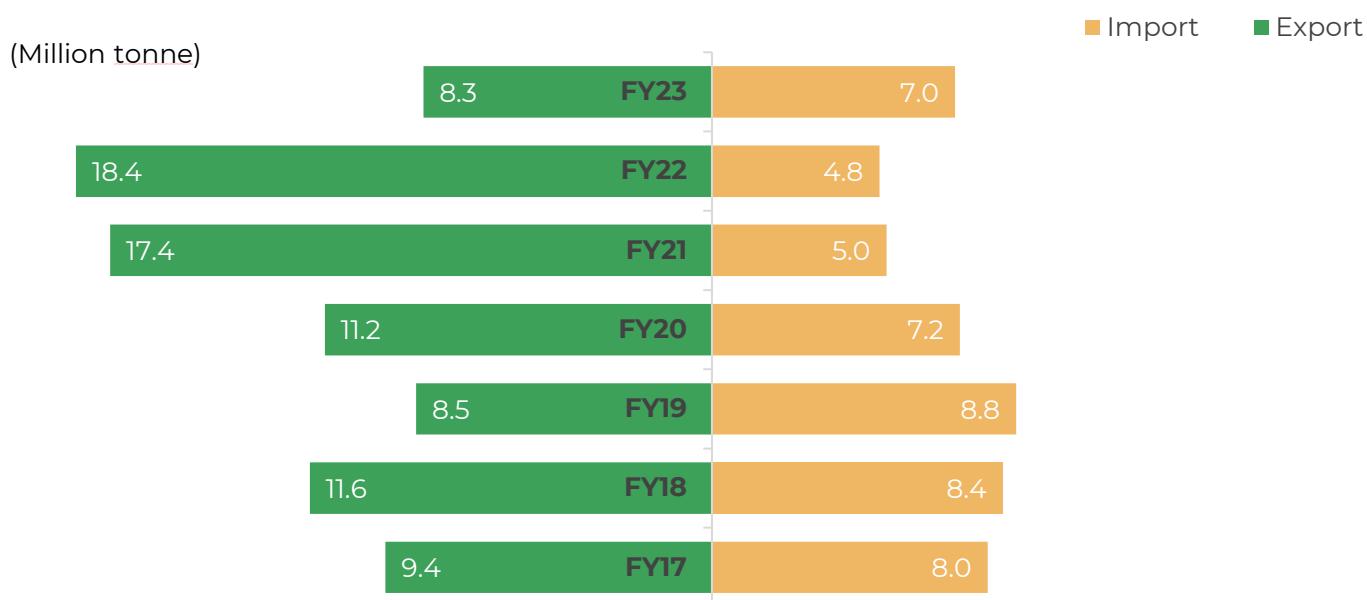
In fiscal 2023, India imported 7 million tonnes of steel, of which 1 million tonne was semi-finished steel and 6 million tonne was finished steel. Of the 6 million tonnes of finished steel, 93% was in the flat category and 7% in the long category.¹⁸ The key importers are Korea, China, and Japan. Due to competitive steel

¹⁸ The gazette of India, Production linked incentive scheme (PLI) for specialty steel in India

prices in Vietnam and Russia, imports increased more than threefold from these destinations in fiscal 23.

To promote India as a 'value-added' steel exporting nation, reduce dependency on imports for specialty steel and in line with the National Steel Policy, 2017 (NSP 2017), the Government of India has launched the Production Linked Incentive (PLI) scheme for the steel sector with a budgetary outlay of Rs 6,322 crore as incentives.¹⁹ This scheme will benefit integrated steel producers as well as micro, small and medium enterprises (MSMEs). The scheme is expected to attract an additional investment of Rs 39,600 crore in the setting up of new plants for producing specialty steel, which has the potential to create an additional capacity of ~25 million tonne in the country for downstream steel manufacturing.²⁰

Figure 12- India's steel imports and exports



Source: Joint Plant Committee, Ministry of Steel, 5

Unit: Million tonne

¹⁹ Ministry of steel, Booklet on Production linked incentive scheme (PLI).

²⁰ Ministry of steel, Booklet on Production Linked Incentive scheme (PLI)

4. Government interventions

NSP 2017:²¹ This policy is an effort to steer the industry to achieve its full potential and enhance steel production with a focus on high-end value-added steel while being globally competitive.

Vision: To develop a technologically advanced and globally competitive steel industry that promotes economic growth.

Key targets:

- Self-sufficiency in steel production by providing policy support & guidance to private manufacturers, MSME steel producers, CPSEs & encourage adequate capacity additions.
- Cost-efficient production and domestic availability of iron ore, coking coal, and natural gas
- Increase per capita steel consumption to 160 kg by fiscal 2031
- To meet the entire demand for high-grade automotive steel, electrical steel, special steel, and alloys for strategic applications domestically by fiscal 2031
- Increase domestic availability of washed coking coal to reduce import dependence on coking coal from ~85% to ~65% by fiscal 2031

PLI scheme:¹³ The objective of the PLI scheme for specialty grade steel is to promote the manufacturing of such grades locally. This incentive will boost the domestic production of specialty steel and attract significant investments in this space. It will also help the Indian steel industry mature in terms of technology and move up the value chain.

In 2021 Production Linked Incentive (PLI) Scheme for Specialty Steel was approved by Union Cabinet with a five-year financial outlay of Rs. 6322 Crore to promote the manufacturing of Specialty Steel. In 2022 67 applications from 30 companies were selected with committed investment of Rs. 42500 Crore.

Steel Import Monitoring System (SIMS):²² This monitoring system was instituted by the Ministry of Commerce and Industry to provide advance information about steel imports to the government and relevant stakeholders. The purpose of the licencing system is to provide statistical data on steel imports entering India prior to their arrival.

Steel and Steel Products (Quality Control) Orders: These orders ensure standardisation of the quality of steel available in the market and safeguard the Indian steel market from substandard imports.

A total of 3,285 Indian Trades Classification (Harmonized System) codes of steel are covered as of fiscal 23, under 152 heads, in the quality control orders on steel.²³

Steel Scrap Recycling Policy (2019):²⁴ The policy was introduced to facilitate and promote the establishment of metal scrapping centres. The scrapping policy ensures that quality scrap is available for the steel industry.

The objective of the policy is to promote a formal and scientific collection, dismantling and processing activities for end-of-life products that are sources of recyclable (ferrous, non-ferrous, and other non-metallic) scraps which will lead to resource conservation and energy savings and setting up of an environmentally sound management system for handling ferrous scrap.

²¹ [The gazette of India, The National Steel Policy 2017](#)

²² [Ministry of steel, Steel Import Monitoring System \(SIMS\): Sample form and Frequently Asked Questions \(FAQs\)](#).

²³ [Ministry of Steel, the gazette of India, Steel and Steel Products \(Quality Control\) Order, 2020](#)

²⁴ [Ministry of steel, the gazette of India, Steel Scrap Recycling Policy](#)

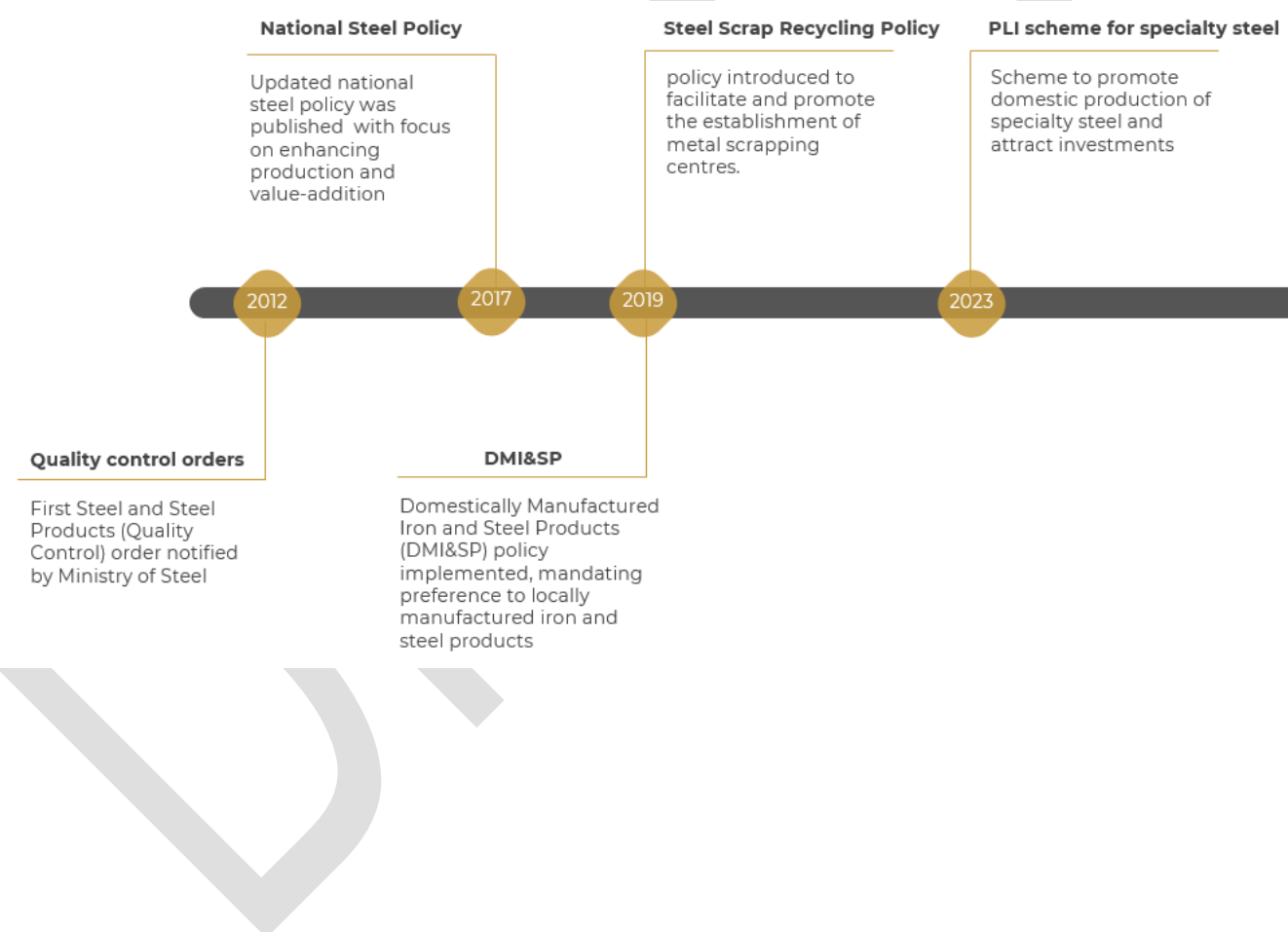
Domestically Manufactured Iron and Steel Products (DMI&SP) policy:²⁵ The policy mandates preference to locally manufactured iron and steel products with a minimum of 15–50% value addition in government procurement. This also supports value-added steel production.

The policy is envisaged to promote growth and development of domestic steel Industry and reduce the inclination to use, low quality low cost imported steel in Government funded projects.

Pilot projects for use of Green Hydrogen in Steel Sector:²⁶ Under the National Green Hydrogen Mission (NGHM), Ministry of New and Renewable Energy (MNRE) will implement pilot projects in the Steel Sector, for replacing fossil fuels and fossil fuel-based feedstock with Green Hydrogen and its derivatives.

The scheme envisages that considering the higher costs of green hydrogen at present, steel plants could begin by blending a small percentage of green hydrogen in their processes, and increasing the blending proportion progressively, with improvement in cost-economics and advancement of technology. The Scheme will be implemented with a total budgetary outlay of Rs. 455 crores till FY 2029-30.

Figure 13 – Major policy timeline for steel manufacturing sector



²⁵ Policy for providing preference to domestically manufactured Iron & Steel Products in Government procurement, Ministry of Steel, the gazette of India, G.S.R. 385(E), Extraordinary, Dated 29th May 2019. And amendment, the gazette of India, G.S.R. 1(E), Extraordinary, Dated 31st December 2020.

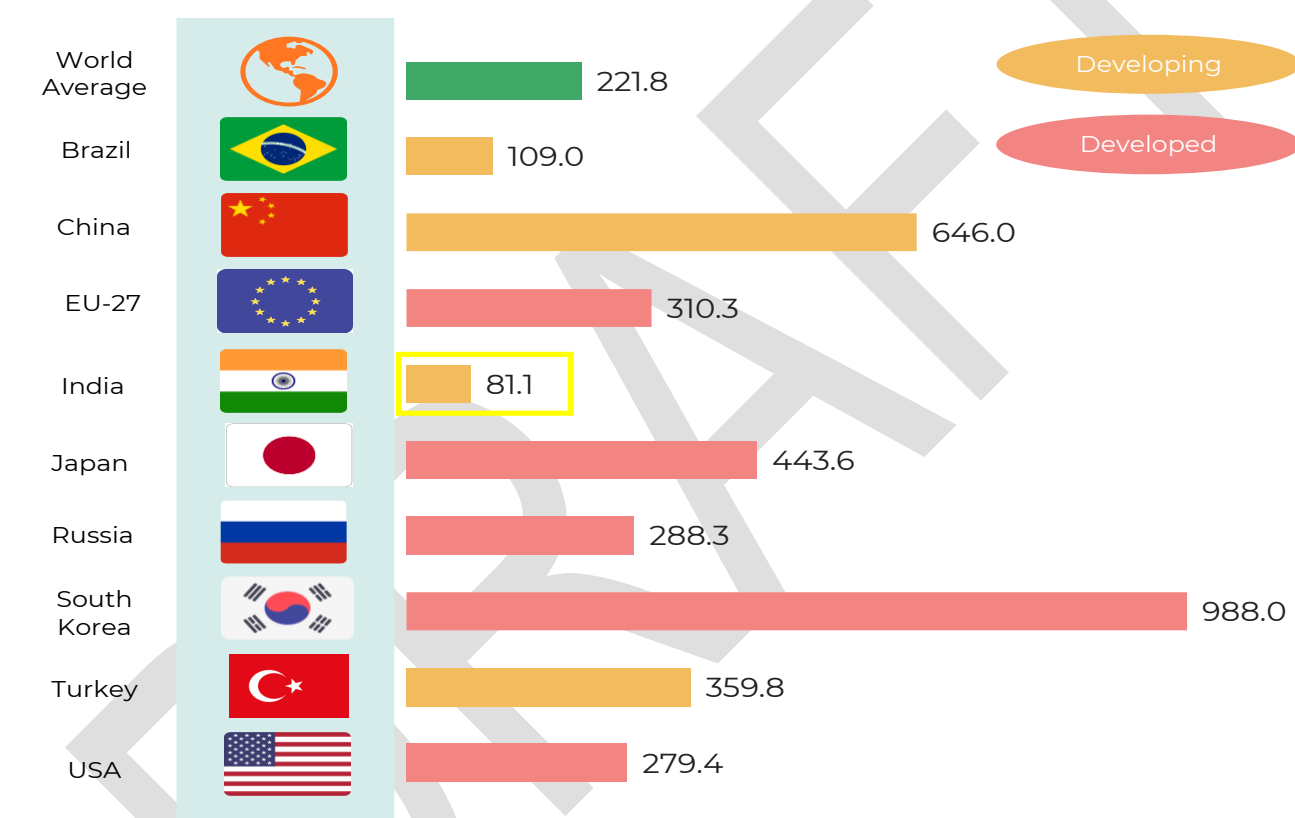
²⁶ Scheme Guidelines for implementation of Pilot projects for use of Green Hydrogen in Steel Sector under the National Green Hydrogen Mission (NGHM), Ministry of New and Renewable Energy, Dated 2nd February 2024.

5. Opportunities and challenges

A. Opportunities: Robust demand from allied sectors to be well supported by consistent supply from domestic players

Growing economy: The Indian economy continues its upward growth trajectory with a focused push on the infrastructure and construction sectors. Several government initiatives such as affordable housing, the expansion of the railway network, development of the domestic shipbuilding industry, opening up of the defence sector for private participation, anticipated growth in the automobile sector, and the boost to the manufacturing sector with the Make in India initiatives, will continue to create significant demand for steel from the domestic sectors.

Figure 13 - Per capita consumption of steel in 2022



Source: World Steel Association, 6

Unit: Kg per person

Export potential: Demand growth is well supported by the supply side, with major Indian producers expanding capacity. This will not only support the domestic steel sector, but also help steel mills utilise India's strategic location and boost export volumes to steel-consuming western nations, while reducing its dependency on the import of value-added steel products. Considering the competitive advantage of steel production in India, the country also has the potential to export steel and become a major player in the global market, thus mitigating the foreign exchange risk emanating from exposure to the global raw materials market, especially for coking coal. To encourage export opportunities and be competitive, the Government of India is contemplating port-led development of steel clusters under the aegis of the Sagarmala programme.

Over the longer run, China is expected to reduce capacity as well as production in line with its government's long-term policy. This will eventually lead to a decline in exports from China, which have already fallen significantly since the 2015–2016 peaks. This presents a ripe opportunity for India to expand its share in the global steel trade, especially in developing markets of Africa and South America. Focus on

the 'Brand India' label will further boost the reputation of Indian-origin steel globally.

Government spending in infrastructure development: The Government of India increased aggregate budgetary support for capex expenditure (capex) by over 28% in fiscal 2024 over FY23RE to Rs. 18.6 lakh crore²⁷. For steel-heavy infrastructure sectors, it has increased 17% to 12.6 lakh crore. The focus is also on urban infrastructure in Tier 2 and 3 cities. Additionally, 100 critical transport infrastructure projects for last-mile and first-mile connectivity for ports, coal, steel, fertiliser, and food grains have been identified. An investment of Rs 75,000 crore has been allocated for these projects, out of which Rs 15,000 crore will come from the private sector. This will improve last-mile and first-mile connectivity. Coupled with Gati Shakti and the National Logistics Policy, this initiative can provide a fillip to the Indian logistics sector and lower logistics costs. As the Indian steel sector leaps towards its aspirational target of 300 million tonne per annum capacity by 2030, such initiatives that lead to demand generation, better connectivity and a lower cost of transportation will help keep the industry growth rate healthy.

B. Major challenges for the steel sector

Availability of coking coal: With steel demand growing at a significant pace, the BF-BOF combination is proven to produce steel economically at this scale. However, this route requires coking coal, majority of which is imported. This dependency makes steel manufacturing vulnerable to global price fluctuations, geopolitical relations, and supply chain disruptions. Though efforts are being made to substitute imported coking coal with domestic coking coal, technological challenges in coal beneficiation and the quality of Indian coal remain concerns.

Emission control: Steel making is one of the hard-to-abate sectors for greenhouse gas emissions. India's emissions of ~2.5 tonne of CO₂ per tonne of crude steel (tCO₂/tcs) are much higher than the global average of 1.91 tCO₂/tcs, primarily due to the poor availability of scrap for furnace charge mix in the production of crude steel.^{14 & 15} This will restrict Indian steel makers' ability to match global emission benchmarks and compete in developed markets. It makes them vulnerable to carbon taxes and non-tariff barriers imposed by developed nations.

Policy and legal issues: There have been few setbacks for steel manufacturing in the past 15 years such as cancellations of coal block allocations in 2014. Blanket bans on iron ore mining in Karnataka (2011-2013) and Goa (2012 - 2014), and subsequent iron ore export bans, have impacted iron ore and steel production in the region. Export duties on iron ore and finished steel in 2022 impacted the operations of export-oriented steel and allied raw material manufacturing facilities. Delays in land acquisition and the availability of land for compensatory afforestation and rehabilitation led to multiple projects facing significant cost and time overruns. These issues are particularly challenging for large-scale manufacturing facilities requiring large land parcels.

High cost of logistics: The steel industry requires ~3 tonne of raw materials to produce 1 tonne of steel from iron ore; this makes logistics cost an important parameter. All the major manufacturers are dependent on railways for iron ore transport. However, the railways have been performing the dual role of functioning as a commercial undertaking and a provider of public utility service, involving the cross-subsidisation of passenger services by freight revenues.²⁸ Indian rail freight rates for iron ore are 500% higher than those of its competitor, Australia.²⁹ Land acquisition issues keep iron ore slurry pipeline penetration low. Though the road network in India has improved in recent years, the non-inclusion of diesel in the Goods and Services Tax (GST) regime leads to increased mining and transport costs for steel manufacturing.

²⁷ Ministry of Finance, The Union Budget for FY 2023-24

²⁸ Comptroller and Auditor General of India, Union Government (Railways) Railways Finances Report - Cross-Subsidization of Passenger and other Coaching Services

²⁹ Ministry of Railway, Research designs and standards organisation, Study of transportation of iron ore over Indian railways.

6. Way forward

Robust growth backed by a green agenda

India is one of the few places in the world with a growing steel manufacturing industry. Robust demand, potential for a higher consumption rate on a low base, policy support for manufacturing capacity growth and the availability of iron ore at relatively low cost are fuelling the steel sector's growth.

Demand 	India's steel demand will remain robust over the next decade with continued infrastructure development, concretisation of houses in rural India, growing urbanisation levels supported by a growing population and nuclearisation of families. Per capita consumption of 81 kg as against the world average of 222 kg indicates potential for rapid growth in the next decade.³⁰ Demand will also be buoyed by rising penetration of automobiles as well as consumer appliances. The Ministry of Steel expects demand to reach 230 million tonnes by fiscal 2031.²¹ The PLI scheme will boost demand from the industrial segment
Supply 	India will need continuous capacity additions to cater to burgeoning demand. Leading integrated steel players are already paving the way with large-scale capex across the country. As utilisation levels continue to hover around the 80% mark, more capacities will come up. India will lead the capacity additions at a global level, which will not only cater to domestic demand, but global as well.² It will help push the Ministry of Steel's target of raising the share of exports, as stated in NSP 2017, and reiterated on multiple occasions thereafter
Decarbonization 	Steel players are undertaking massive capacity expansions without abandoning their decarbonisation agenda. They are adopting the best available technologies across the spectrum to reduce GHG emissions and meet their near-term targets. Several players teamed up with various European technology partners to accelerate the process. As per the Nationally Determined Contributions (NDCs) for the Indian steel industry submitted to the Ministry of Environment, Forestry and Climate Change, the average CO₂ emission intensity of the Indian steel industry was projected to reduce from 3.1 tCO₂/tcs in 2005 to 2.64 tCO₂/tcs by 2020 and 2.4 tCO₂/tcs by 2030. With the efforts of all stakeholders involved, specific CO₂ emissions will be reduced to ~2.5 tCO₂/tcs in 2020.^{13&14} Aided by the ministry's focus on improving domestic scrap availability, primary and secondary steel makers aim to increase scrap offtake even as ironmaking remains the primary source for steelmaking.

30 [World Steel Association, 2023 World Steel in Figures](#)

21 [The gazette of India, The National Steel Policy 2017](#)

2 [Ministry of Steel, Joint Plant Committee, Annual statistics 2016-17, 2017-18, 2021-22, 2022-23](#)

13 [Ministry of steel, Energy & environment management in steel sector](#)

14 [World Steel Association, Released 2022, Sustainability Indicators 2022 report](#)

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NITI Aayog